
Morphology-Agnostic Animation-to-Robot Motion Retargeting via Sceneflow-Conditioned Diffusion

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Abstract

Humanoid motion retargeting has advanced considerably, but existing pipelines are largely confined to mappings between humans and humanoid robots—asking a real robot to imitate the vivid choreography of an animated character remains effectively impossible. We formalize this gap as Morphology-Agnostic Animation-to-Robot Motion Retargeting: given an *arbitrary animation video* and the kinematic specification of a target humanoid, the goal is to produce a physically plausible joint-level trajectory that reproduces the source motion on the target robot. Unlike prior feature-matching retargeting methods, we leverage **sceneflow** to disentangle motion information across diverse animations and drive the target robot. Specifically, we propose **FlowOp**, which encodes sceneflow through a dual-branch visual skeleton encoder, binds it to the target skeleton via geometry-driven proximity attention, and decodes joint trajectories with a rectified-flow Diffusion Transformer—producing *physically plausible motion* directly executable on the target robot. To support this task, we curate **AnimBot**, a benchmark of 7,800+ paired animation-to-robot motion clips with a manually audited cross-morphology split. **[Jiahao: Our method achieves an MPJPE of 0.77 on same-morphology videos and 5.62 on the cross-morphology stylized-character benchmark]**., demonstrating that decoupling motion content from source morphology provides a principled foundation for animation-to-robot transfer. Videos and demos are available at our anonymous project page: xxx.

1 Introduction

Humanoid robots are transitioning from controlled laboratories to complex real-world environments, where acquiring diverse motor skills remains fundamental. The dominant paradigm leverages large-scale human motion data—videos and motion-capture databases—to guide robot policies via imitation or reinforcement learning [1, 2]. **Motion retargeting** bridges human demonstrations and robotic execution: classical approaches solve joint configurations via inverse kinematics and differential optimization [3], while learning-based methods fit implicit mappings from paired data [4, 5].

Despite their progress, these methods are largely confined to retargeting between humans and humanoid robots, and typically require explicit knowledge of the humanoid’s body proportions and a hand-designed correspondence between human and robot joints. Recent generative approaches have begun to extend motion retargeting beyond the human-only setting: MoCapAnything [6, 7] predicts skeletal motion for arbitrary rigged 3D assets, and Motion 3-2-4 [8] reconstructs 4D motion from a video and a paired source asset. These methods face two key limitations: (i) they remain confined to *same-species* mappings, where source and target share compatible body structure, and (ii) they fail to produce *physically plausible* joint trajectories deployable on real robots.

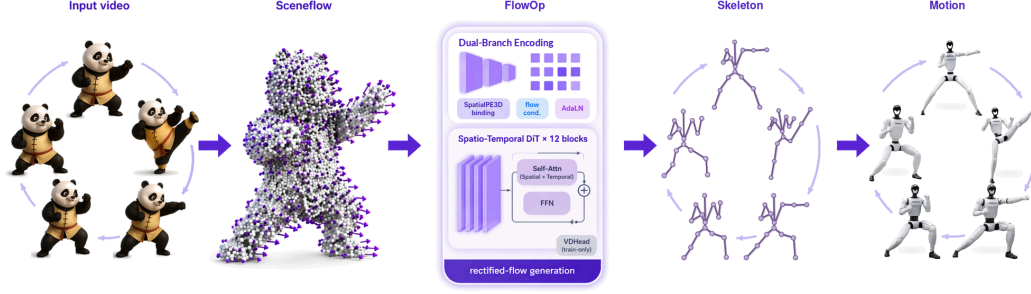


Figure 1: **Overview of FlowOp.** Given an arbitrary animation character, FlowOp reconstructs sceneFlow from the input video and infers a morphology-agnostic skeleton sequence. Conditioned on a reference humanoid robot, the model predicts physically plausible, deployable motion through an end-to-end framework, enabling the robot to imitate the animated character.

We address this gap by formalizing a new task: **Morphology-Agnostic Animation-to-Robot Motion Retargeting (MA-A2R)**. Given an *arbitrary animation video*—spanning humans, animals, and stylized characters (Fig. 3)—and the kinematic specification of a target humanoid (e.g., joint hierarchy, link topology, and joint limits), the goal is to drive the robot directly from the video, without any pre-defined source-to-target body correspondence.

The key obstacle to MA-A2R is that conventional motion descriptors—joint angles, mesh vertices, or SMPL parameters—are all tied to a specific source body. Once the body changes, these representations no longer carry meaningful information. Our key insight is to represent motion as a *spatial displacement field* instead: **sceneFlow**, the per-point 3D displacement of the observed surface over time. Because sceneFlow is indexed by spatial location rather than source-point identity, the same action, like forward motion, turning, or limb swing, produces comparable displacement patterns regardless of whether the source is a human or a stylized character.

Building on this observation, we propose **FlowOp**, a sceneFlow-conditioned video-to-robot motion transfer framework, as illustrated in Fig. 1. Raw sceneFlow is first processed by a dual-branch encoder that separates surface geometry from motion, producing per-point features in 3D space. To bridge source motion and target morphology, a Skeleton-Anchored Attention module binds each target joint to its spatial neighborhood through geometric proximity in a shared canonical frame—sidestepping the need for any semantic joint correspondence like [7]. Conditioned on these bound features, a spatio-temporal Diffusion Transformer trained under rectified-flow generation decodes the final joint trajectories. During training, a Visual Dynamics Head further regularizes the model by enforcing causal consistency between predicted motion and future visual states.

Because the conditioning signal is motion-centric rather than source-structured, the resulting transfer is *retargeting-free* in a precise source-side sense: (i) no hand-designed source-to-target joint correspondence is used; (ii) no per-source-category priors or skeleton-matching rules are involved; and (iii) the source animation is never first converted into a human or category-specific pose space. Geometric binding is learned end-to-end from sceneFlow, not from a correspondence ontology.

To support this task, we create **AnimBot**, a benchmark of 7,800+ paired animation-to-robot motion clips spanning diverse stylized characters, together with a manually-audited cross-morphology split for evaluation under stylized re-skinning. **[Jiahao: Proposed FlowOp reaches MPJPE 0.77 on same-morphology video and 5.62 on the cross-morphology stylized-character benchmark, outperforming MoCapAnything by $9.3\times$ and $5.8\times$ respectively].**

In summary, our contributions are threefold: (i) we formalize **MA-A2R**, a new task that drives a humanoid robot from arbitrary animation videos without source-to-target joint correspondence; (ii) we propose **FlowOp**, the sceneFlow-conditioned architecture for skeleton-anchored motion generation, integrating dual-branch encoding, geometric Skeleton-Anchored Attention, and a rectified-flow Diffusion Transformer with Flow-Augmented AdaLN and a training-only Visual Dynamics Head; (iii) we release **AnimBot**, a benchmark of 7,800+ paired animation-to-robot clips with a manually-audited cross-morphology split. Extensive experiments validate that FlowOp outperforms state-of-the-art baselines on both same- and cross-morphology settings.

75 2 Related Work

76 **Motion Retargeting.** Motion retargeting transcribes source-side motion onto a target skeleton.
 77 Classical methods use constrained motion transformation with physical filtering or shape-aware
 78 adaptation [9, 10]. In humanoid robotics, optimization-based approaches align human and robot
 79 poses through IK and Cartesian constraints [11, 12], with PHC’s SMPL-based pipeline becoming a
 80 common preprocessing path [13, 14, 2]. OmniH2O extends this to whole-body teleoperation [15],
 81 and GMR addresses artifacts such as foot sliding and ground penetration [3]. Learning-based methods
 82 seek reusable correspondences from data: NKN [16] uses forward kinematics with cycle consistency,
 83 Skeleton-Aware Networks [17] operate convolutions over kinematic trees, and SAME [18] learns
 84 skeleton-agnostic motion embeddings. For humanoid robots, NMR [4] and IKMR [5] learn supervised
 85 mappings from human to robot motion distributions. All these methods, however, assume compatible
 86 source-target morphologies and rely on explicit or implicit joint correspondences—assumptions that
 87 break in the MA-A2R setting, where the source body is entirely unknown.

88 **Video-to-Humanoid Motion Learning.** A recent line of work couples monocular video directly
 89 with humanoid policy learning: HumanPlus [1] performs real-time shadowing of human motion and
 90 learns a humanoid policy supervised on a large-scale human motion corpus; OmniH2O [15] applies a
 91 multi-modal teacher-student framework to whole-body teleoperation; HOVER [2] unifies multiple
 92 humanoid control interfaces under a single policy; MeshMimic [19] reconstructs human motion
 93 and terrain geometry from video for motion-terrain learning. Heracles [20] is adjacent to this line,
 94 inserting a state-conditioned diffusion layer between reference motions and low-level trackers. These
 95 systems share a design choice well suited to their intended setting: the policy’s action space is directly
 96 aligned with humanoid kinematics, taking full advantage of the in-family prior that exists when the
 97 source is human and the target is humanoid. Our work targets a different setting: in MA-A2R, the
 98 source is an arbitrary animation character, which need not admit a faithful human-pose interpretation.

99 **Video-based Motion Reconstruction and Generation.** Human-oriented methods regress paramet-
 100 ric body parameters from video to obtain frame-level motion [21, 22]. Broader-category approaches
 101 relax source assumptions: MoCapAnything [6, 7] predicts 3D joint trajectories of a target skeleton
 102 from video, and Motion 3-2-4 [8] reconstructs motion via per-vertex displacements on a reference
 103 mesh. These methods are fundamentally feature-mapping based—each image feature is tied to a
 104 source-side body model (SMPL vertex, skeleton joint, or mesh vertex)—and motion is recovered
 105 through this correspondence. In MA-A2R, however, large morphological gaps leave many source
 106 features *without matching counterparts*, causing such methods to fail. We instead represent motion
 107 via scene-flow, which encodes how points move in 3D rather than where they map on a body model,
 108 sidestepping dependence on source morphology and enabling direct transfer to real robots.

109 3 Method

110 **Overview.** This section presents FlowOp under the MA-A2R setting. We first formalise the
 111 task and clarify its boundary with conventional retargeting in Sec. 3.1, then motivate scene-flow as
 112 the core motion descriptor in Sec.3.2, and detail the framework: source-agnostic 3D observation,
 113 geometry-driven cross-morphology binding, and rectified-flow DiT generation in Sec.3.3.

114 3.1 Task Formulation

115 MA-A2R (Morphology-Agnostic Animation-to-Robot Motion Retargeting) drives a humanoid robot
 116 directly from arbitrary animation videos. Given monocular video $\mathcal{V} = \{I_t\}_{t=0}^T$ with any source
 117 character (human, animal, or stylized) and target skeleton $\mathcal{S}$, MA-A2R generates motion sequence
 118 $\mathbf{M}_{1:T}$ that reproduces the source motion, without requiring source skeleton, mesh, or body attributes.

119 3.2 Why MA-A2R Needs Scene-flow

120 The core distinction between MA-A2R and conventional retargeting lies in source-side assumptions.
 121 Traditional methods anchor visual features to source-body components (skeleton joints, mesh vertices),
 122 relying on structural correspondence that fails when source morphology is unknown. We therefore
 123 explore three motion representations that do not require feature-to-body correspondence: **3D tracking**

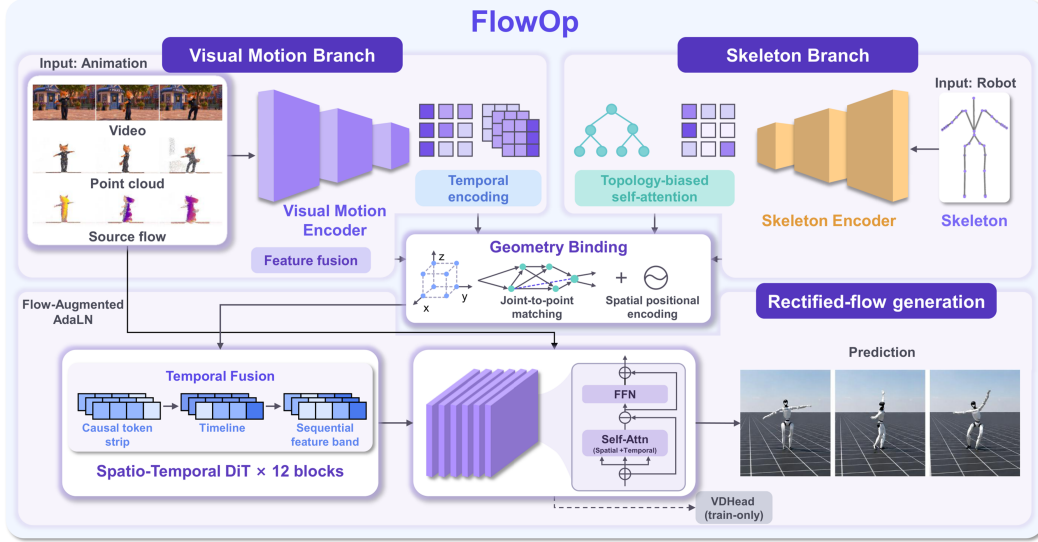


Figure 2: **FlowOp architecture.** *Visual Motion Branch* extracts motion features from point cloud and sceneflow. *Skeleton Branch* encodes target kinematic structure. *Geometry Binding* aligns the two via 3D-proximity-based attention. *DiT Decoder* generates joint trajectories under rectified-flow, conditioned on binding features, frames history, and mean sceneflow direction (Flow-Augmented AdaLN). *Visual Dynamics Head* (VDHead) provides training-time supervision.

records surface point motion indexed by persistent identity, tying displacement to source surface layout. Point identities have no stable counterpart across morphologies, and occlusion breaks tracking continuity. **Optical flow** provides dense image-plane motion but remains a 2D field—depth-axis displacement must be inferred indirectly, and the signal is entangled with camera geometry and texture. **Sceneflow** assigns 3D displacement to observed surface points. Its primary variable is motion itself, not point identity or image projection. The same action—forward progression, turning, limb swing—produces comparable displacement patterns regardless of source morphology, providing MA-A2R a morphology-agnostic 3D motion field that binds directly to the target skeleton.

3.3 Framework

FlowOp comprises three stages, illustrated in Fig. 2: (i) extracts sceneflow and anchor trajectory from video (Sec. 3.3.1); (ii) binds visual motion to target skeleton via 3D proximity (Sec. 3.3.2); (iii) decodes joint trajectories with a rectified-flow DiT augmented by Flow-Augmented AdaLN and Visual Dynamics supervision (Sec. 3.3.3).

3.3.1 Visual Observation and Global Anchoring

Unlike methods [8, 6] that operate on raw RGB frames, FlowOp first rewrites the monocular animation into a 3D observation. We use Track4World [23] to estimate world-frame point trajectories, and use Depth Anything 3 [24] as the depth backbone to lift pixels into a unified world coordinate system. After stitching temporal chunks, we retain N foreground tracks and obtain point positions $\mathbf{P} \in \mathbb{R}^{(T+1) \times N \times 3}$ and sceneflow $\mathcal{F} \in \mathbb{R}^{T \times N \times 3}$, where T denotes the number of inter-frame intervals and $\mathcal{F}_{t,n}$ is the 3D displacement of point n from frame t to $t+1$ ($t \in [0, T-1]$, $n \in [1, N]$). This representation is intentionally physical and source agnostic: downstream modules see only geometry and motion, not source skeletons, body models, or semantic keypoints.

To reduce learning burden and remove residual morphology dependence, we recover the global trajectory analytically. Let $\bar{\mathcal{F}}_t = \frac{1}{N} \sum_{n=1}^N \mathcal{F}_{t,n}$ be the spatial mean sceneflow at frame t , approximating the subject’s bulk translational velocity. We initialise the trajectory with the median of the [Jiahao: first-frame foreground points, $\mathbf{h}_0 \in \mathbb{R}^3$], and accumulate mean sceneflow:

$$\mathbf{h}_0 = \text{median}_{n=1}^N \mathbf{P}_{0,n}, \quad \mathbf{a}_t = \mathbf{h}_0 + \sum_{\tau=0}^{t-1} \bar{\mathcal{F}}_\tau \quad (1)$$

[Jiahao: *where $\mathbf{h}_0$ is the initial anchor*], $\mathbf{P}_{0,n}$ is the world coordinate of point n at frame 0, $\bar{\mathcal{F}}_\tau$ is the mean scene-flow at interval τ , and $\mathbf{a}_t$ is the global anchor at frame t . This makes $\mathbf{a}_{0:T}$ a deterministic function of $(\mathbf{P}, \mathcal{F})$ alone, eliminating SMPL fitting, keypoint detection, and skeleton alignment. As a result, source agnosticism is no longer a learned behavior but a hard representational constraint. Source morphologies span orders of magnitude in scale. We normalize all spatial inputs by the rest-skeleton bounding-box diagonal. [Jiahao: *what is rest-skeleton, need definition*].

3.3.2 Dual-Branch Encoding with Geometric Binding

After the 3D observation step in Sec. 3.3.1, the animation is represented by point positions $\mathbf{P}$ and scene-flow $\mathcal{F}$. We next rewrite this source-side point motion as target-side joint conditions. The generator can then predict motion directly on the robot skeleton.

Visual motion branch. The visual branch encodes point-level 3D motion from the animation, the centered coordinate $\mathbf{P}_{t,n} - \bar{\mathbf{P}}_t$ together with scene-flow $\mathcal{F}_{t,n}$. Feature fusion and temporal encoding then yield point features $\mathbf{F}_{\text{vis}} \in \mathbb{R}^{T \times N \times D}$, where D is the feature dimension. We deliberately retain all N foreground points rather than pooling them early, since each target joint must subsequently query a distinct local region in 3D space.

Skeleton branch. The skeleton branch describes the target robot. Given rest-pose joints $\mathbf{J}^{\text{rest}} \in \mathbb{R}^{K \times 3}$ and the kinematic tree, it applies tree message passing. Topology-biased self-attention then yields feature $\mathbf{Q}_{\text{skel}} \in \mathbb{R}^{K \times D}$, where K is the number of robot joints. This feature carries joint identity, bone geometry, and topology.

Geometry binding bridges the visual and skeleton branches by converting point-indexed motion features into joint-indexed conditions. Because MA-A2R provides no reliable source skeleton or semantic body parts, predefined source-to-target joint correspondences are unavailable; instead, we let *each robot joint actively query* its own neighbourhood of source points in a shared canonical 3D space. To this end, target joints are centered at the pelvis $\mathbf{J}_0^{\text{rest}}$ and visual points at the per-frame foreground centroid $\bar{\mathbf{P}}_t$: $\mathbf{J}^c = \mathbf{J}^{\text{rest}} - \mathbf{J}_0^{\text{rest}}$, $\mathbf{P}_{t,n}^c = \mathbf{P}_{t,n} - \bar{\mathbf{P}}_t$. Both coordinates use the same spatial positional encoding $\psi(\cdot)$ [Jiahao: *$\psi(\cdot)$ sinusoidal? Fourier? Is Attn cross-attention or self-attention? what is Key and value here?*].:

$$\mathbf{F}_{\text{bnd}} = \text{Attn}(\mathbf{Q}_{\text{skel}} + \psi(\mathbf{J}^c), \mathbf{F}_{\text{vis}} + \psi(\mathbf{P}^c)). \quad (2)$$

The initial binding feature is $\mathbf{F}_{\text{bnd}} \in \mathbb{R}^{T \times K \times D}$. Through this attention, each robot joint gathers scene-flow evidence from its nearby source points. The representation changes from source-point-indexed motion to target-joint-indexed conditioning.

Temporal fusion. Since the binding step yields one feature per target joint per time step, we further apply causal temporal fusion along each joint sequence to capture short-term dynamics, producing $\mathbf{F}_{\text{bnd}}^{\text{tmp}} \in \mathbb{R}^{T \times K \times D}$. This feature preserves the local visual motion around each joint and serves as the main visual condition for the DiT generator.

3.3.3 Generation: Rectified-Flow DiT with FA-AdaLN and VD Head

With joint-level visual conditions, the generator no longer operates on source points. It predicts anchor-relative motion in the target joint space. We use a 12-layer spatio-temporal DiT as the generative backbone. Its token sequence combines $\mathbf{F}_{\text{bnd}}^{\text{tmp}}$, the history token $\mathbf{H}_{\text{hist}}$, and the noisy motion token $\mathbf{X}_s$. The history token $\mathbf{H}_{\text{hist}}$ comes from recently generated frames and keeps motion continuous. The motion token $\mathbf{X}_s$ lies between random noise and the clean motion $\mathbf{X}_1 \in \mathbb{R}^{T \times K \times 4}$. [Jiahao: *What is 4 here? quaternion? root+rotation?*]. We add $\mathbf{Q}_{\text{skel}}$ to matching joint tokens, so the DiT sees robot topology at every step.

Rectified-flow training follows x_1 -prediction [Jiahao: *x_1 or X_1 ?*]. We sample $\mathbf{X}_s$ between Gaussian noise and the clean anchor-relative motion. The DiT then predicts $\hat{\mathbf{X}}_1$ from $\mathbf{X}_s$ and the conditioning signals. These signals are $\mathbf{F}_{\text{bnd}}^{\text{tmp}}$, $\mathbf{Q}_{\text{skel}}$, $\mathbf{H}_{\text{hist}}$, and a global-flow condition.

Flow-Augmented AdaLN. While Geometry Ginding already provides local evidence around each robot joint, it lacks an explicit global motion trend; mean scene-flow, in contrast, better captures forward direction, translation speed, and turning. FA-AdaLN therefore injects this complementary signal as a per-frame global-flow condition $\mathbf{g}_t^{\text{flow}} = W_{\text{flow}} \bar{\mathcal{F}}_t$, where $W_{\text{flow}} \in \mathbb{R}^{D \times 3}$ maps 3D mean

sceneflow to the DiT hidden dimension. Rather than being appended as an extra token, this condition enters the AdaLN pathway of the noisy motion tokens, modulating the normalization parameters inside each DiT block. With this conditioning, denoising is written as

$$\hat{\mathbf{X}}_1 = f_\theta(\mathbf{X}_s, s; \mathbf{F}_{\text{bnd}}^{\text{tmp}}, \mathbf{Q}_{\text{skel}}, \mathbf{H}_{\text{hist}}, \mathbf{g}_t^{\text{flow}}). \quad (3)$$

Thus, $\mathbf{F}_{\text{bnd}}^{\text{tmp}}$ constrains local joint motion. The flow condition supplies global movement. Together they guide stable robot trajectory generation. **[Jiahao: What is s here? time step?]**.

During training, we add a Visual Dynamics Head (VDHead) as an auxiliary regularizer that reads intermediate DiT hidden states and predicts future visual dynamics. This encourages the model to link current robot motion with later visual changes while discouraging shortcuts based on single-frame co-occurrence. VDHead is removed at inference and thus adds no generation cost.

Training Objective. Since end effectors are the most sensitive to cross-morphology motion fidelity, we weight the primary x_1 reconstruction loss by kinematic depth: **[Jiahao: $\hat{\mathbf{X}}_{1,t,k}$? $\mathbf{X}_{1,t,k}$?]**.

$$\mathcal{L}_{x1} = \mathbb{E}_{s,\epsilon} \left[\sum_{t=1}^T \sum_{k=1}^K (1 + \beta \text{depth}(k)) \left\| \hat{\mathbf{X}}_{1,t,k} - \mathbf{X}_{1,t,k} \right\|_1 \right], \quad (4)$$

where $\text{depth}(k)$ is the depth of joint k in the robot kinematic tree. The scalar β controls the end-effector weighting strength. The expectation is over rectified-flow time s and Gaussian noise ϵ . The complete objective is

$$\mathcal{L} = \mathcal{L}_{x1} + \lambda_{\text{vd}} \mathcal{L}_{\text{vd}} + \mathcal{L}_{\text{aux}}, \quad (5)$$

where $\mathcal{L}_{\text{vd}}$ is the VDHead regularizer. The term $\mathcal{L}_{\text{aux}}$ contains physical auxiliary losses. Appendix C gives their weights and design.

3.3.4 Inference: From Video to Robot Motion

Inference follows the same data path: the video first yields $\mathbf{P}$, $\mathcal{F}$, $\bar{\mathcal{F}}$, and the clip-level anchor $\mathbf{a}_{0:T}$, from which the dual-branch encoder and Geometry Binding produce $\mathbf{F}_{\text{bnd}}^{\text{tmp}}$ and $\mathbf{Q}_{\text{skel}}$; conditioned on these features, the DiT then samples the anchor-relative motion $\hat{\mathbf{X}}_1$ from random noise.

For long videos, we use sliding windows of 256 frames with 32-frame overlap, blended by a Hann window: the first window is initialized with a standing rest pose as history, while subsequent windows reuse the last 8 generated frames from the previous one. Since $\mathbf{a}_{0:T}$ is computed at the clip level, all windows share a single global path, avoiding translation breaks across window boundaries.

Since the DiT output is anchor-relative, we recover world coordinates by combining it with the global anchor $\mathbf{a}_t$ from Sec. 3.3.1. Letting the decoded motion contain a root offset $\mathbf{r}_t$ and pelvis-relative joint offsets $\{\mathbf{q}_{t,k}\}_{k=1}^K$, the pelvis position $\mathbf{p}_{t,0}$ and the remaining joint positions $\mathbf{p}_{t,k}$ are

$$\mathbf{p}_{t,0} = \mathbf{a}_t + \mathbf{r}_t, \quad \mathbf{p}_{t,k} = \mathbf{p}_{t,0} + \mathbf{q}_{t,k}. \quad (6)$$

The anchor thus carries the global displacement while the DiT supplies the local robot pose, and their combination yields the joint trajectories used for tracking and deployment.

3.4 Downstream Deployment: Tracking and Sim-to-Real

To validate physical executability, we train a general-purpose motion tracker [25] that converts FlowOp-generated kinematic trajectories into torque commands. The tracker uses a motion-latent policy pretrained on diverse corpora and fine-tuned on FlowOp outputs, with standard tracking rewards and domain randomization. Successful deployment (Fig. 5) on real hardware shows that MA-A2R trajectories are dynamically coherent and physically executable. Details in Appendix F.

4 Experiments

4.1 Datasets and Evaluation

Benchmark. To support MA-A2R, we construct **AnimBot**, a benchmark of 7,800+ paired animation-to-robot motion clips (Fig. 3). We first retarget AMASS motions onto the target robot

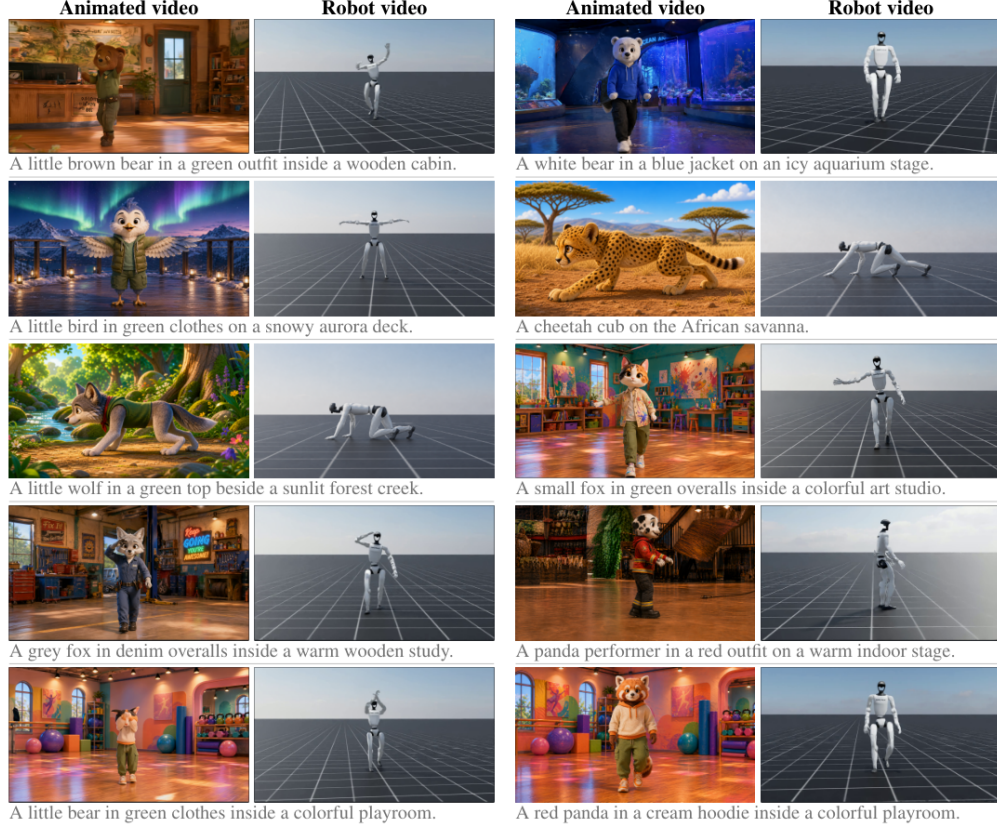


Figure 3: **AnimBot benchmark gallery.** Each pair shows a stylized animated video (**left**) and the corresponding *deployable* robot motion (**right**), with the text prompt used for animation generation below. Animation drives the model; robot motion serves as ground truth.

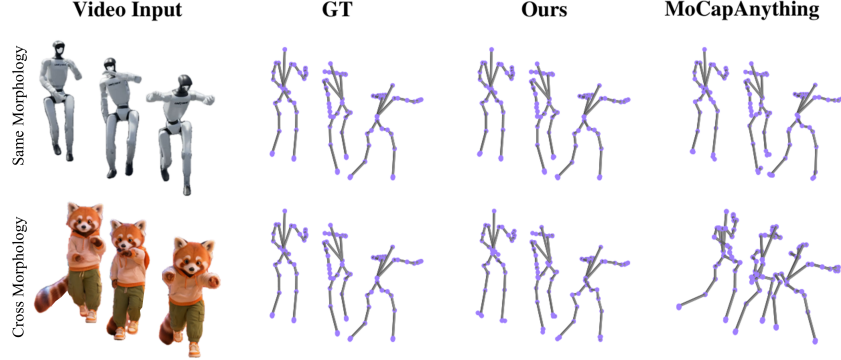


Figure 4: **Qualitative skeleton comparison.** From left to right: input frames, ground truth, FlowOp, and MoCapAnything V2, under same-morphology (**top**) and cross-morphology (**bottom**) settings.

238 skeleton to obtain ground-truth trajectories, then apply Wan2.2 [26] re-skinning on rendered robot
 239 videos to produce stylised animations spanning diverse characters. After manual auditing, 395
 240 motion-aligned clips are retained as a gold evaluation set for testing motion fidelity under substantial
 241 appearance and morphology shift. Construction details are in Appendix B.

242 **Evaluation metrics.** We report MPJPE, MPJVE, and CD-Skeleton to evaluate relative pose accu-
 243 racy, temporal consistency, and skeleton-structure fidelity. And ablations instead report RAW-MPJPE
 244 and PA-MPJPE in physical millimetres, which separate absolute error, global trajectory drift, and
 245 local pose-shape error. Full definitions are in Appendix D.



Figure 5: **Animation-to-robot transfer in the real world.** Source animations (**top**) and the resulting motions deployed on a real humanoid (**bottom**).

Table 1: Main results under same-morphology (robot-rendered) and cross-morphology (Wan2.2-restyled animation) settings, evaluated on the same 395 motion-aligned clips. Lower is better; all numbers are averaged over 3 seeds.

Method	Same-Morphology			Cross-Morphology		
	MPJPE ↓	MPJVE ↓	CD-Skel ↓	MPJPE ↓	MPJVE ↓	CD-Skel ↓
MoCapAnything [6]	7.14	0.80	0.23	32.39	1.87	0.44
FlowOp (ours)	0.77	0.11	0.01	5.62	0.40	0.13

4.2 Training Procedure

We train FlowOp with a two-stage curriculum on eight A100 GPUs with bfloat16 mixed precision. Stage 1 pre-trains the dual-branch encoder for 50 epochs with a lightweight 2-block scaffold DiT, supervised only by $\mathcal{L}_{x1}$ and a trajectory loss; the under-capacity scaffold concentrates gradient pressure on the encoder. Stage 2 runs for 100 epochs with the full 12-block DiT and the complete objective ($\mathcal{L}_{x1} + \mathcal{L}_{vd} + \mathcal{L}_{aux}$); the dual-branch encoder is frozen for the first 10 epochs and then jointly fine-tuned. Optimisation uses AdamW with cosine decay and per-group learning rates (3×10^{-4} for DiT, 3×10^{-5} for dual-branch encoder). Training details, like full hyperparameters, are in Appendix E.

4.3 Main Results

Same-morphology comparison. On 395 held-out robot-rendered clips, where source and target share morphology, FlowOp consistently outperforms MoCapAnything [6] (Table 1): MPJPE drops from 7.14 to 0.77 ($9.3\times$), MPJVE from 0.80 to 0.11 ($7.3\times$), and CD-Skeleton from 0.23 to 0.01 ($> 20\times$). These gains confirm that scene-flow-conditioned generation more accurately recovers relative pose, velocity continuity, and skeletal structure.

Cross-morphology comparison. On AnimBot, the same 395 motions are re-rendered as stylised animation characters, keeping motion fixed while appearance and morphology change substantially. Table 1 shows wider gap under this shift: FlowOp reaches MPJPE 5.62 vs. 32.39 ($5.8\times$), MPJVE 0.40 vs. 1.87 ($4.7\times$), and CD-Skeleton 0.13 vs. 0.44 ($3.4\times$). Feature-mapping reconstruction collapses under source-side change, while FlowOp’s scene-flow factors out appearance, feeding the generator 3D motion indexed by spatial location rather than source-body features. The smaller MPJVE ratio does not imply better tracking by MoCapAnything: its predictions often drift toward near-static motion, where velocity error collapses to the ground-truth magnitude itself. Qualitative results in Fig. 4 show that MoCapAnything loses the core motion structure, whereas FlowOp retains low position error and stable dynamics. Notably, FlowOp’s error grows by $7.3\times$ from same- to cross-morphology, demonstrating strong motion preservation under substantial appearance shift.

Deployment. Using the tracker in Sec. 3.4, FlowOp-generated trajectories transfer faithfully to a real Unitree G1 across diverse stylized animations (Fig. 5), confirming that MA-A2R outputs are dynamically coherent and physically executable.

4.4 Ablation Study

Descriptor analysis: from outcomes to mechanisms. We isolate the descriptor’s contribution from two angles, summarised in Table 2: a reconstruction study (*whether* scene-flow helps) and a

Table 2: **Descriptor analysis.** *Left:* Descriptor-isolated reconstruction (RAW-MPJPE in mm; FA-AdaLN and VD Head disabled). *Right:* Diagnostic scores on 300 held-out clips (higher is better; Uniformity is undefined for 2D optical flow). All descriptors use $N_{\text{points}}=4096$.

Descriptor	Reconstruction		Diagnostics		
	Spatial Dim.	RAW-MPJPE ↓	$\widehat{A}_g \uparrow$	$\widehat{C}_p \uparrow$	$U_n \uparrow$
3D Tracking	3D (identified trajectories)	434.20	0.97	0.03	0.70
Optical Flow	2D (image-plane)	502.20	0.99	0.00	–
Sceneflow (ours)	3D (surface-volume)	152.60	0.98	0.75	0.80

Table 3: Stage-2 module ablation, same-morphology. Errors are reported in millimetres.

Method	RAW-MPJPE ↓	PA-MPJPE ↓
Ours w/o FA, w/o VD	152.60	82.30
Ours w/o VD	94.20	80.50
Ours w/o FA	67.20	17.40
Ours	20.20	10.00

277 diagnostic study (*why*). FA-AdaLN and VD Head are disabled, and all descriptors use $N_{\text{points}}=4096$;
 278 full setup is in Appendix A.

279 *Reconstruction.* Training three identical models with only the descriptor changed, sceneflow reaches
 280 152.60 mm RAW-MPJPE, far below 3D tracking (434.20 mm) and optical flow (502.20 mm). Fig. 6
 281 in Appendix reveals the failure modes: optical flow exhibits depth collapse—it specifies where
 282 pixels move, not how the body moves in 3D—while 3D tracking preserves rough structure but
 283 produces incoherent motion with freezing and identity-locked artifacts, organising motion as persistent
 284 trajectories rather than a continuous field.

285 *Diagnostics.* The three diagnostic scores explain these failures. Optical flow attains the highest
 286 Agnosticism (0.99) but *zero* Completeness, consistent with the rank-deficient projection that loses
 287 depth-axis motion. 3D tracking is reasonably agnostic (0.97) but weak in Completeness (0.03) and
 288 Uniformity (0.70): information is organised by persistent identities, leaving off-template surface
 289 motion under-represented. Sceneflow achieves the best Completeness (0.75) and Uniformity (0.80)
 290 while staying near the top in Agnosticism (0.98)—a balance that translates directly into the lower
 291 joint error observed above.

292 **Module ablation.** We isolate the two key generator modules on the same-morphology test set
 293 (Table 3): **FA** (Flow-Augmented AdaLN) injects spatially-averaged sceneflow into the DiT AdaLN
 294 condition for global motion guidance, and **VD** (Visual Dynamics Head) regularises causal dynamics
 295 by predicting future visual states during training. The descriptor-only baseline (w/o FA, w/o VD)
 296 yields 152.60 mm RAW-MPJPE; adding FA alone halves RAW-MPJPE to 94.20 mm but leaves
 297 PA-MPJPE high at 80.50 mm, indicating that FA anchors global trajectory alignment but does little
 298 for local pose. VD alone produces the opposite pattern—PA-MPJPE drops sharply from 82.30 to
 299 17.40 mm—confirming its role in local skeletal fidelity. Enabling both yields 20.20 mm / 10.00 mm,
 300 far beyond either single module. This complementarity is intuitive: VD shapes a physically plausible
 301 local motion manifold on which FA’s global alignment lands more cleanly, while FA’s stable reference
 302 frame frees VD to focus on internal kinematics—global guidance and local calibration jointly produce
 303 FlowOp’s accurate, physically coherent motion.

304 5 Conclusion

305 We introduced **MA-A2R**, a task that drives a humanoid robot from *arbitrary* animation videos
 306 without predefined source-to-target joint correspondence. Our key insight is to decouple motion from
 307 source identity via **sceneflow**, a spatially-indexed displacement field comparable across radically
 308 different bodies. Proposed **FlowOp** binds sceneflow to the target skeleton via geometric attention
 309 and decodes joint trajectories with a rectified-flow DiT, consistently outperforming prior methods on
 310 our **AnimBot** benchmark across both same- and cross-morphology settings. These results establish
 311 morphology-agnostic descriptors with skeleton-anchored decoders as a principled foundation for
 312 cross-body motion transfer.

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A Full Descriptor Analysis and Diagnostics

This appendix contains the original descriptor-space analysis, the formal transfer criteria, the detailed failure-mode argument for 3D tracking, optical flow, and scene flow, the optical-flow rank-deficiency proof, and the three-criteria diagnostic design behind Table ?? . The empirical results remain in the main text; the appendix supplies the formal and experimental details needed to interpret them.

A.1 Signal Families

We restrict our analysis to descriptors that expose motion directly from video, without first collapsing it into a source-specific skeleton or mesh space. Three families of visual motion signals dominate this design space:

- **3D point tracking**: estimate temporally consistent trajectories of queried surface points or pixels, often in world coordinates;
- **Optical flow**: estimate a dense image-plane motion field that assigns a 2D displacement to each pixel between frames;
- **Scene flow**: estimate pointwise 3D motion, i.e., a 3D displacement vector for each observed point between frames.

All three are motion-bearing visual signals extracted from video without first imposing source-body semantics. This is exactly why they form the right comparison queue for our task. The key question is not whether they contain motion, since they all do, but whether they preserve motion in a form that remains meaningful after the source morphology is discarded. For video-to-robot transfer, the useful content is not the identity of a particular source landmark, but body-independent action structure such as heading change, forward progression, contact rhythm, bilateral phase, limb swing timing, and speed profile.

A.2 Transfer Criteria

Let $\mathcal{V}$ denote a video observation of an arbitrary moving subject, and let $\mathbf{M}^*$ denote the underlying *motion semantics*, namely the body-independent kinematic content that any articulated target should reproduce, such as stride frequency, heading change, and interaction phases. A **motion descriptor** is a mapping

$$\Phi : \mathcal{V} \longrightarrow \mathcal{X} \quad (7)$$

that extracts from the raw video a representation $\Phi(\mathcal{V}) \in \mathcal{X}$ intended to serve as the conditioning signal for a downstream motion generation model $g : \mathcal{X} \times \mathcal{S} \rightarrow \mathbf{M}$, which produces motion $\mathbf{M}$ for a target skeleton $\mathcal{S}$. For cross-morphology transfer to be well defined, Φ must satisfy three properties.

Criterion 1 (Subject-Agnosticism). Φ should retain as little information as possible about the source subject’s identity ID and morphology Morph. Formally, we measure

$$\text{Ag}(\Phi) := 1 - I(\Phi(\mathcal{V}); \text{ID}, \text{Morph}) / H(\text{ID}, \text{Morph}) \quad (8)$$

with I the mutual information and H the entropy. Higher $\text{Ag}(\Phi) \in [0, 1]$ indicates stronger decoupling between motion and the source body. When Φ leaks morphology, the downstream generator g is forced to undo the coupling, a task that generally lacks a well-defined inverse for morphologies different from the source.

Criterion 2 (Motion Completeness). Φ should preserve all motion semantics needed to reconstruct $\mathbf{M}^*$. Formally,

$$\text{Cp}(\Phi) := I(\Phi(\mathcal{V}); \mathbf{M}^*) / I(\mathcal{V}; \mathbf{M}^*) \in [0, 1] \quad (9)$$

If Φ is a data-processing bottleneck that destroys motion information in $\mathcal{V}$, no downstream generator can recover it. By the data-processing inequality, $\text{Cp}(\Phi) = 1$ is achievable only when Φ is a sufficient statistic for $\mathbf{M}^*$ in $\mathcal{V}$.

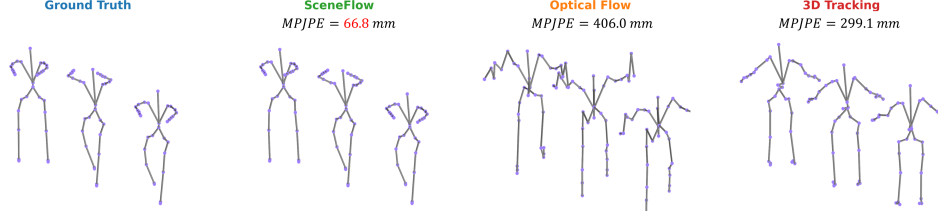


Figure 6: **Generated motion under different descriptors.** Optical flow collapses in depth, while 3D tracking shows identity-locked artifacts. SceneFlow preserves a coherent 3D skeleton structure closely matching the ground truth.

Criterion 3 (Spatial Uniformity). Φ should distribute its support uniformly over the moving subject, so that every body region is observed with comparable signal density. Pure voxel occupancy is insufficient: a category-specific tracker can achieve the same point-budget count as a uniform sampler while concentrating all its support on anatomical landmarks, leaving muscle, skin, and non-template regions unobserved. Let $\{\mathcal{B}_1, \dots, \mathcal{B}_R\}$ be a partition of the subject volume into R semantically meaningful regions (head, torso, left/right upper arm, left/right lower arm, left/right upper leg, left/right lower leg), and let $p_r(\Phi) = |\text{supp}(\Phi) \cap \mathcal{B}_r|/|\text{supp}(\Phi)|$ be the fraction of descriptor support in region r . We measure spatial uniformity by the Shannon entropy of this distribution, normalized to $[0, 1]$:

$$\text{Un}(\Phi) := -\frac{1}{\log R} \sum_{r=1}^R p_r(\Phi) \log p_r(\Phi) \in [0, 1], \quad (10)$$

A uniformly distributed descriptor attains $\text{Un} = 1$; a fully concentrated descriptor ($p_r = 1$ for a single region) attains $\text{Un} = 0$. Uniformity is necessary because the geometric binding we use in the downstream architecture (§3) matches each target joint to its nearest source support points. If an entire target region has no nearby descriptor support, that binding degenerates and no downstream training signal can repair it.

A descriptor that simultaneously maximizes Ag , Cp , and Un is a sufficient, body-agnostic, uniformly supported motion representation, the kind of signal cross-morphology generation requires.

The information-theoretic quantities in Eqs. (8) and (9) serve as conceptual desiderata. Directly estimating them for high-dimensional video descriptors is intractable in our setting. We therefore evaluate defect-targeted diagnostics $\widehat{\text{Ag}}$ and $\widehat{\text{Cp}}$ in Section ?? . Failure on these tests is evidence that the corresponding desideratum is violated, while strong performance indicates better alignment with it. Spatial Uniformity is measured directly via Eq. (10) once the body regions are specified.

A.3 Descriptor Failure Modes and the Case for SceneFlow

All three descriptors carry motion, but they package it through different variables and therefore different nuisance factors. This distinction matters because the downstream robot generator never sees the source body directly; it only sees the descriptor. The question is which descriptor carries motion in the least morphology-bound form while still preserving the 3D information needed for transfer.

3D Tracking (Φ_{trk}). A 3D tracking pipeline outputs a set of N identified trajectories

$$\Phi_{\text{trk}}(\mathcal{V}) = \{ \mathbf{T}_i(t) \in \mathbb{R}^3 \mid i = 1, \dots, N, t \in [0, T] \} \quad (11)$$

where each trajectory carries an *identity index* i that anchors the i -th point to a specific visual feature of the source subject throughout time. These trajectories certainly contain motion: if a forelimb swings, the torso advances, or the body turns, the tracked paths record that evolution over time. The problem is how this motion is packaged. Formally, Φ_{trk} is indexed by persistent identity $i \in \{1, \dots, N\}$, whereas sceneFlow is indexed by spatial location $\mathbf{X} \in \mathbb{R}^3$. Identity indices do not survive morphology change; spatial locations do.

For cross-morphology transfer, two tracker families are relevant. Template-bound systems such as 4DHumans and WHAM anchor outputs to human-specific structure; generic point trackers such as

CoTracker and TAPIR relax the category prior but still output identity-persistent trajectories. In both cases, the motion signal remains tied to which source support point moved. That identity coupling injects source morphology directly into the descriptor, because the semantic unit is not “forward motion happened here” but “track i moved this way.” Template-bound trackers compound the issue by concentrating support around skeleton joints or landmark-like regions, which lowers Spatial Uniformity and under-represents surface motion away from the template. Generic trackers reduce the category prior, but not the central issue for our setting: the descriptor is still indexed by persistent source-point identities and therefore remains less transferable once the source body is discarded.

Optical Flow (Φ_{of}). Optical flow provides a dense 2D displacement field $\Phi_{\text{of}}(u, v) \in \mathbb{R}^2$ at every image pixel. It therefore also carries motion: swing timing, local speed variation, and image-plane direction changes are all visible in the flow field. Its weakness is that it carries apparent motion in the image plane rather than the transferable 3D motion required by the target robot.

Depth ambiguity. Optical flow is the projection of the underlying 3D scene flow onto the image plane. For a point $\mathbf{X}$ with 3D velocity $\mathbf{v}$ imaged by a pinhole camera, the optical flow at its image projection is

$$\mathbf{u}_{\text{of}} = \mathbf{J}_{\pi}(\mathbf{X}) \cdot \mathbf{v}, \quad \mathbf{J}_{\pi}(\mathbf{X}) \in \mathbb{R}^{2 \times 3} \quad (12)$$

where $\mathbf{J}_{\pi}$ is the projection Jacobian. Because $\mathbf{J}_{\pi}$ has rank 2, its kernel is a one-dimensional subspace along the viewing ray, and all 3D velocities differing by a motion along this ray are indistinguishable under Φ_{of} (see Section A.4 for the full derivation). Thus, 3D motion components along the viewing ray are not identifiable from instantaneous monocular optical flow without additional geometric or temporal priors. Any downstream model that needs full 3D motion, which any articulated target driven in 3D does, must reconstruct this missing component indirectly, yielding $\text{Cp}(\Phi_{\text{of}}) < 1$ whenever axial motion carries task-relevant semantics.

Projection dependence. Optical flow is measured in pixels, not metres. Its magnitude depends on field of view and subject distance: the same physical displacement produces a larger vector when the subject is close and a smaller one when it is far. Consequently, the descriptor mixes motion semantics with camera geometry. This does not make optical flow useless as a motion cue, but it does mean that the same underlying action is encoded differently under different viewpoints and scales, weakening its suitability as a morphology-independent transfer signal.

Appearance dependence. Optical flow relies on photometric pixel correspondence. Untextured regions, the aperture problem, and self-occlusion, such as arms crossing the torso, cause track drift, spurious flow, and holes. The descriptor therefore entangles motion with appearance: highly textured areas produce strong flow, while untextured areas produce noise. This introduces a subject-specific appearance signal into what should be a pure motion representation.

The net effect is that optical flow is a dense image-space descriptor. It undeniably contains motion, but not in the fully 3D, camera-robust form that cross-morphology robot transfer needs.

Sceneflow (Φ_{sf}). Sceneflow assigns a 3D displacement vector to every point of a dynamic point cloud:

$$\Phi_{\text{sf}}(\mathbf{X}) = \frac{d\mathbf{X}}{dt} \in \mathbb{R}^3 \quad (13)$$

Sceneflow carries motion through the 3D displacement itself. If a body advances, turns, compresses, or swings a limb, that kinematic event appears directly as metric 3D motion on the observed surface. This changes the representation bias in a crucial way: the primary variable is no longer which source point moved, as in tracking, or how motion projected to the image plane, as in optical flow, but what 3D motion occurred. As a result, actions performed by morphologically different bodies can still share similar displacement-level structure, including forward progression, heading change, contact rhythm, bilateral phase, or limb-swing timing, even when their exact joints, meshes, and proportions differ.

Sceneflow does not eliminate geometry altogether; the observed surface still has shape. But among the three descriptors considered here, it is the one whose motion content is least entangled with source-specific support identities and camera projection. That makes it the strongest candidate for Subject-Agnosticism. It also retains the most direct access to the 3D displacement field underlying $\mathbf{M}^*$, making it the strongest candidate for Motion Completeness. Finally, because it is defined over

observed surface points rather than sparse landmarks, its support is substantially more even across the body, making it the strongest candidate for Spatial Uniformity.

The central point is therefore simple: sceneflow contains motion information in the most morphology-independent form among the video-derived descriptors we compare. This is exactly the bias the video-to-robot setting requires, and it is why sceneflow is our preferred descriptor before any downstream architecture is introduced.

A.4 Optical Flow Rank Deficiency

We detail the claim in Section A.3 that optical flow systematically loses motion information along the camera viewing ray.

Setup. Let a 3D point $\mathbf{X} = (X, Y, Z)^\top$ in the camera frame move with velocity $\mathbf{v} = \dot{\mathbf{X}} \in \mathbb{R}^3$. Under a standard pinhole camera with focal lengths (f_x, f_y) and principal point (c_x, c_y) , the image projection is

$$\pi(\mathbf{X}) = \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} f_x X/Z + c_x \\ f_y Y/Z + c_y \end{pmatrix}. \quad (14)$$

The optical flow at the projected pixel is the time derivative of $\pi(\mathbf{X})$:

$$\mathbf{u}_{\text{of}} = \frac{d}{dt}\pi(\mathbf{X}) = \mathbf{J}_\pi(\mathbf{X})\mathbf{v}, \quad \mathbf{J}_\pi(\mathbf{X}) = \frac{1}{Z} \begin{pmatrix} f_x & 0 & -f_x X/Z \\ 0 & f_y & -f_y Y/Z \end{pmatrix}, \quad (15)$$

Proposition 1 (Viewing-ray nullspace of optical flow). *For any point $\mathbf{X}$ with $Z > 0$, the Jacobian $\mathbf{J}_\pi(\mathbf{X}) \in \mathbb{R}^{2 \times 3}$ has rank exactly 2, and its one-dimensional right null space is spanned by $\mathbf{X}$ itself:*

$$\ker \mathbf{J}_\pi(\mathbf{X}) = \text{span}\{\mathbf{X}\} = \{\lambda \mathbf{X} : \lambda \in \mathbb{R}\}. \quad (16)$$

Consequently, two 3D velocities $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^3$ produce identical optical flow if and only if $\mathbf{v}_1 - \mathbf{v}_2 \in \text{span}\{\mathbf{X}\}$, i.e., they differ by a translation along the viewing ray.

Proof. Direct substitution yields $\mathbf{J}_\pi(\mathbf{X})\mathbf{X} = \frac{1}{Z}(f_x X - f_x X, f_y Y - f_y Y)^\top = \mathbf{0}$, so $\mathbf{X} \in \ker \mathbf{J}_\pi$. To see the null space is one-dimensional, observe that the two rows of $\mathbf{J}_\pi$ are linearly independent whenever $Z > 0$ (the leading 2×2 block is diagonal with positive entries $f_x/Z, f_y/Z$), so $\text{rank } \mathbf{J}_\pi = 2$ and by rank-nullity $\dim \ker \mathbf{J}_\pi = 1$. The equivalence follows because $\mathbf{J}_\pi(\mathbf{v}_1 - \mathbf{v}_2) = \mathbf{0}$ iff $\mathbf{v}_1 - \mathbf{v}_2 \in \ker \mathbf{J}_\pi$. $\square$

Consequence for motion descriptors. Proposition 1 implies that motion components along the viewing ray are not identifiable from *instantaneous monocular optical flow* without additional geometric or temporal priors. In the Completeness measure of Eq. (9), this yields a strict inequality $\text{Cp}(\Phi_{\text{of}}) < 1$ whenever depth-axis motion carries relevant semantics. Our axial-completeness diagnostic in Section A.5.3 tests this missing component directly in the descriptor pipeline used in the paper. Sceneflow, which directly observes $\mathbf{v} \in \mathbb{R}^3$ without applying $\mathbf{J}_\pi$, has no such rank deficiency and preserves all three velocity components.

A.5 Empirical Protocols and Three-Criteria Diagnostics

This subsection records the experimental details behind the two descriptor studies in Section ?? . The reconstruction study tests whether a descriptor can support downstream joint recovery under identical model capacity. The diagnostic study then asks which theoretical defect explains the observed gap. All descriptors are sub-sampled to a matched point budget $N=4096$ whenever a point budget is applicable, ruling out support-count effects.

Descriptor instantiation. All three descriptors are extracted from the same source video stream. *Sceneflow* is recovered from the dynamic point cloud sequence. *Optical flow* is estimated per consecutive frame pair via WAFT on the rendered RGB stream, and N image-plane locations are sub-sampled per frame. The *category-specific 3D tracker* follows the output profile of template-bound human trackers (4DHumans, WHAM): at each frame we take the detected skeleton joint positions and sample N points from an isotropic Gaussian centered at each joint, with variance calibrated so that per-joint point clouds just meet at mid-limb. This reproduces the template-bound behavior relevant to our comparison.

A.5.1 Descriptor-Isolated Reconstruction Protocol

To isolate the effect of the motion descriptor as much as possible, we train the identical FlowOp architecture three times. The only change is the descriptor stream and the minimal descriptor-specific lifting needed to express each input in the common point-token interface consumed by the SceneEncoder. The three variants are: (i) identity-persistent 3D tracking trajectories, where the model receives how tracked source points move over time; (ii) optical flow, densely sampled at $N=4096$ image-plane locations and lifted only to match the input interface; and (iii) scene-flow, represented as observed-surface points with metric 3D displacements. The lift used for optical flow supplies comparable support coordinates, but it does not restore the missing depth-axis component of the 2D flow descriptor.

All three models share the same training corpus, training schedule, and 150-epoch budget. They are evaluated on the same 300 held-out clips and repeated over 3 seeds; the standard deviation of raw-MPJPE is below 15 mm. FA-AdaLN and the VD Head are disabled in all three runs, so the comparison cannot rely on global-flow conditioning or future-dynamics supervision. The reported raw-MPJPE is measured in physical millimetres, before mesh-unit-volume normalisation. This convention exposes absolute structural and trajectory drift directly. The same raw-MPJPE scale is reused in the Stage-2 ablation in Tables 3 and ??; the scene-flow row in Table ?? is the same descriptor-only configuration as *Ours w/o FA, w/o VD* in Table 3.

Under this controlled setup, the error gap in Table ?? is attributed primarily to the descriptor. Scene-flow reduces raw-MPJPE by roughly $3\times$ compared with either alternative. The qualitative comparison in Fig. 6 gives the corresponding failure modes: optical flow collapses along depth and distorts the skeleton because it only describes image-plane displacement, while 3D tracking preserves a rougher metric body structure but exhibits freezing, drift, and identity-locked configurations. Scene-flow provides a continuous surface-level 3D displacement field, giving the generator a cleaner signal for reconstructing joint motion.

A.5.2 Measuring Subject-Agnosticism

Design. We run two complementary leakage tests, each designed to expose one specific Agnosticism failure mode from Section A.3. (i) *View leakage*. We train a Ridge probe that takes the clip-pooled descriptor and predicts the camera-to-subject distance in metres, extracted from the scene’s camera extrinsics. Optical flow, being defined in pixel units whose magnitude scales with $1/\text{distance}$, must encode this quantity linearly; metric-3D descriptors like scene-flow do not. (ii) *Category leakage*. We train a Ridge probe that predicts a $3K$ -dim signature of the GT source skeleton, formed by flattening pelvis-relative joint positions at $t=0$. A template-bound tracker whose support lives on skeleton joints exposes this signature directly in its coordinates; a subject-agnostic descriptor cannot.

Metric. $\widehat{\text{Ag}}(\Phi) = 1 - \frac{1}{2}(R_{\text{view}}^2(\Phi) + R_{\text{cat}}^2(\Phi))$, with both R^2 clipped to $[0, 1]$ and averaged over 5-fold cross-validation; higher is better.

Why this measures what it should. Subject-Agnosticism is $1 - I(\Phi; \text{ID}, \text{Morph})/H$ (Eq. (8)), but we do not attempt to estimate that mutual information directly. Instead, we ask whether nuisance quantities that should not define the action, such as camera distance or skeleton signature, are linearly recoverable from Φ . Positive probe performance is evidence that such nuisance information is exposed by the descriptor. The two probes jointly cover the two leakage channels of interest: optical flow’s projection-geometry leakage and the tracker’s category-template leakage. Restricting the probe to be linear ensures that the diagnostic only rewards information that is directly exposed by the descriptor.

A.5.3 Measuring Motion Completeness

Design. We decompose motion completeness into two tests, each isolating one structural failure mode. (i) *Axial completeness* $\widehat{\text{C}}_{\text{p}\parallel}$. We extract a per-frame descriptor feature Φ_t and train a Ridge probe that predicts *only the depth-axis component* of the ground-truth joint velocity vector. By Proposition 1, optical flow’s depth component is structurally unrecoverable under the instantaneous optical-flow model, so this test is the direct experimental counterpart of the rank-deficiency analysis. (ii) *Surface completeness* $\widehat{\text{C}}_{\text{p}\perp}$. At each frame we sample non-joint surface points on the subject body, specifically midpoints of bones and centroids of surface patches between joints, and train a

probe that predicts their ground-truth velocities from Φ_t . A template-bound tracker has zero support at these locations and therefore cannot recover this signal; sceneflow observes the surface directly and retains it.

Metric. $\widehat{Cp}(\Phi) = \frac{1}{2}(\widehat{Cp}_{\parallel} + \widehat{Cp}_{\perp})$, where each sub-score is $R^2 \in [0, 1]$ under 5-fold cross-validation; higher is better.

Why this measures what it should. Motion Completeness is the fraction of motion information preserved in Φ (Eq. (9)), but again we do not estimate that quantity directly. Instead, the two sub-tests isolate two motion components that any descriptor for this task must preserve: camera-axial motion and non-joint surface motion. Failure on either sub-test is sufficient evidence of incompleteness; strong performance indicates better preservation of motion information relevant to M^* . Decomposing into axial ($\widehat{Cp}_{\parallel}$) and non-joint-surface ($\widehat{Cp}_{\perp}$) tests therefore exposes optical flow’s depth loss and the tracker’s template bias in the exact forms analyzed in Section A.3.

A.5.4 Measuring Spatial Uniformity

Design. We partition each subject’s body into $R=10$ canonical regions (head, torso, left/right upper arm, left/right lower arm, left/right upper leg, left/right lower leg; hands and feet grouped with their adjacent distal limb) by proximity to GT joints. For each frame, we assign every 3D descriptor support point to its closest region, producing a region-wise distribution $p_r(\Phi)$ of the N support points, and report the normalized Shannon entropy of this distribution as defined in Eq. (10). This score is defined only for descriptors whose support lives in the 3D body volume, namely sceneflow and the category-specific 3D tracker. Optical flow supports live in the 2D image plane, where body-region assignments are view-dependent and not a property of the descriptor, so Un is not defined for optical flow and reported as “–” in Table ??.

Metric. $Un(\Phi) \in [0, 1]$: 1 when the N support points are distributed evenly across the 10 regions, and 0 when they are concentrated in a single region. Higher is better.

Why this measures what it should. Spatial Uniformity formalizes the observation from Section A.3 that the tracker’s defect is not low point count but the *semantic bias* of where those points live. Entropy is the classical choice for measuring how concentrated a distribution is, and normalization by $\log R$ places the metric on $[0, 1]$ regardless of the number of regions. A template-bound tracker has p_r concentrated on the joint-hosting regions and yields low entropy; a surface-uniform descriptor like sceneflow has near-uniform p_r and yields entropy close to 1. The metric therefore directly measures the property that Skeleton-Anchored Attention requires for well-posed geometric binding.

Diagnostic interpretation. The unrounded diagnostic scores underlying Table ?? are 0.969/0.027/0.700 for 3D tracking, 0.988/0.000/– for optical flow, and 0.984/0.750/0.801 for sceneflow. Optical flow achieves $\widehat{Cp} = 0$, dominated by $\widehat{Cp}_{\parallel} = 0$. This is the empirical realization of Proposition 1: in the diagnostic setting, the depth component of motion is absent from the descriptor, so no linear probe can recover it. The 3D-tracking baseline achieves only $\widehat{Cp} = 0.027$, dominated by the failure of $\widehat{Cp}_{\perp}$. Its support is organized by persistent or template-like point identities, leaving too little information to reconstruct velocities on off-template surface regions. The same support bias yields the lower Uniformity score among 3D descriptors. Sceneflow simultaneously dominates Completeness, by roughly $28\times$ over the next best descriptor, and Uniformity, while staying close to the best Agnosticism score. This is why the diagnostic table and the reconstruction table point to the same conclusion.

Each of the three diagnostics targets a specific theoretical defect from Section A.3: camera geometry leakage, template leakage, depth ambiguity, non-rigid surface loss, and template bias. A descriptor that scores high on all three diagnostics is a strong candidate for cross-morphology motion generation.

B Benchmark Construction Details

Wan2.2 re-skinning fidelity. A naive application of Wan2.2-Animate would let the diffusion sampler infer pose from its own image-side estimator, producing visually plausible but motion-

shifted clips that are useless as a motion benchmark. We instead drive the generator with the robot ground-truth motion through two modifications: (i) *raw-GT src_pose substitution*, where the official preprocessor’s pose video is replaced by one rendered directly from the held-out robot motion sequence so that every frame of generation is conditioned on the true target motion; (ii) *automatic pose-2D anchoring on the reference image*, where each reference character is run through a pose detector to extract its pelvis pixel coordinate and the rendered `src_pose` is rescaled and translated per clip so that pose tokens land at the right pixel positions for that specific reference figure, preserving the reference character’s proportions instead of drifting toward a generic silhouette. After generation, the restyled candidates were manually audited by the authors for one-to-one motion correspondence with their source robot ground truth. The 395 clips that passed this audit form the gold cross-morphology benchmark; candidates with visible drift, missing limbs, or temporal incoherence are excluded from this held-out evaluation set. These conditioning steps are used only to make Wan2.2 produce motion-faithful restyled benchmark videos; they are not components of FlowOp inference.

C Physical Auxiliary Losses

This appendix records the physical auxiliary losses used in Stage 2. The overall objective combines the primary reconstruction loss $\mathcal{L}_{x1}$, the VD regulariser $\mathcal{L}_{vd}$, and the auxiliary term $\mathcal{L}_{aux}$ defined below. The auxiliary bundle contains trajectory fidelity, root velocity, pelvis-relative joint position, joint velocity, bone-length preservation, ground-penetration suppression, and foot-contact classification:

$$\mathcal{L}_{aux} = \lambda_{traj}\mathcal{L}_{traj} + \lambda_{rv}\mathcal{L}_{root_vel} + \lambda_{pos}\mathcal{L}_{pos} + \lambda_{vel}\mathcal{L}_{vel} + \lambda_{bn}\mathcal{L}_{bone} + \lambda_{pen}\mathcal{L}_{pen} + \lambda_{ct}\mathcal{L}_{contact}. \quad (17)$$

All auxiliary terms are weighted by the rectified-flow time step s , so that physical constraints contribute more strongly at high noise levels. Table 4 lists the loss type and fixed weight used in all experiments. For the kinematic-depth weighting in $\mathcal{L}_{x1}$ (Eq. 4), we use $\beta=0.3$, which gives end effectors at depth 5–7 a weight of $2.5\text{--}3.1\times$ the root.

Table 4: Physical auxiliary losses used in Stage 2. Auxiliary terms are s -weighted.

Term	Type	Weight	Description
$\mathcal{L}_{x1}$	L1	1.0	Primary reconstruction with kinematic-depth weighting
$\mathcal{L}_{vd}$	Cosine	0.05	Visual dynamics anticipation (Eq. ??)
$\mathcal{L}_{traj}$	L1	2.0	World-space root trajectory fidelity
$\mathcal{L}_{root_vel}$	L1	0.3	World-space root velocity
$\mathcal{L}_{pos}$	L1	1.0	Pelvis-relative joint positions
$\mathcal{L}_{vel}$	L1	0.5	Joint velocity and temporal smoothness
$\mathcal{L}_{bone}$	Smooth L1	0.5	Parent-child bone-length preservation
$\mathcal{L}_{pen}$	ReLU	0.2	Ground-penetration suppression
$\mathcal{L}_{contact}$	BCE	0.1	Foot-contact classification

D Evaluation Metrics

This appendix gives the evaluation metrics used in this paper. Main results use a root-aligned mesh-unit-volume scale. We first normalise joint positions by the rest-pose mesh bounding box. We then subtract the pelvis position at every frame. This removes asset-scale differences and global translation error, so the main tables focus on relative pose, temporal consistency, and skeletal structure.

Let K be the number of joints and T the number of frames. Let $\hat{\mathbf{j}}_{t,k}$ and $\mathbf{j}_{t,k}^*$ denote the normalised, root-aligned predicted and ground-truth joint positions. **MPJPE** measures mean per-joint position error:

$$\text{MPJPE} = \frac{1}{TK} \sum_{t=1}^T \sum_{k=1}^K \left\| \hat{\mathbf{j}}_{t,k} - \mathbf{j}_{t,k}^* \right\|_2. \quad (18)$$

MPJVE applies the same measure to first-order temporal differences:

$$\text{MPJVE} = \frac{1}{(T-1)K} \sum_{t=1}^{T-1} \sum_{k=1}^K \left\| \Delta \hat{\mathbf{j}}_{t,k} - \Delta \mathbf{j}_{t,k}^* \right\|_2. \quad (19)$$

701 **CD-Skeleton** is a symmetric skeleton Chamfer distance. It matches each joint to the closest parent-
 702 child bone segment in the opposite skeleton:

$$\text{CD-Skel} = \frac{1}{2} \left(D(\hat{\mathcal{J}}, \mathcal{B}^*) + D(\mathcal{J}^*, \hat{\mathcal{B}}) \right), \quad (20)$$

703 where $\mathcal{J}$ is a joint set and $\mathcal{B}$ is a set of bone segments. The point-to-segment distance is

$$D(\mathcal{J}, \mathcal{B}) = \frac{1}{|\mathcal{J}|} \sum_{\mathbf{p} \in \mathcal{J}} \min_{(\mathbf{a}, \mathbf{b}) \in \mathcal{B}} \text{dist}(\mathbf{p}, \overline{\mathbf{ab}}). \quad (21)$$

704 The main tables report MPJPE and MPJVE multiplied by 100 for readability. Ablations use RAW-
 705 MPJPE in physical millimetres, which exposes absolute structural and trajectory error. We also report
 706 PA-MPJPE after per-frame Procrustes alignment. The gap between RAW-MPJPE and PA-MPJPE
 707 indicates whether error comes mainly from global drift or local skeletal shape.

708 E Training Details

709 We train FlowOp with a two-stage curriculum. All experiments use eight NVIDIA A100 GPUs with
 710 bfloat16 mixed precision. Optimisation uses AdamW with a 5-epoch linear warmup followed by
 711 cosine decay. The DiT and SceneEncoder are assigned separate parameter groups with learning rates
 712 3×10^{-4} and 3×10^{-5} , respectively; this ratio is fixed across all experiments. Reported results are
 713 averaged over three random seeds.

714 **Stage 1: SceneEncoder pre-training.** Stage 1 pre-trains the SceneEncoder for 50 epochs. We
 715 attach a lightweight 2-block scaffold DiT and supervise it only with the primary reconstruction loss
 716 $\mathcal{L}_{x1}$ and a cumulative trajectory loss. The scaffold is deliberately under-capacity: it cannot replace
 717 the encoder as the motion modelling module, so the main gradient pressure is concentrated on the
 718 SceneEncoder.

719 **Stage 2: Joint DiT training.** Stage 2 runs for 100 epochs. The scaffold is replaced by the full
 720 12-block DiT, and training uses the complete rectified-flow objective, the VD regulariser, and the
 721 physical auxiliary losses. The SceneEncoder is frozen for the first 10 epochs, allowing the DiT to
 722 adapt to the pretrained feature space; it is then unfrozen and trained jointly with the DiT at the lower
 723 learning rate.

724 **Time-step sampling.** The rectified-flow time step s is sampled from a mixture of 80% Logit-
 725 Normal and 20% Uniform over $[0, 1]$. This emphasizes the mid-noise regime while keeping endpoint
 726 coverage.

727 **History conditioning.** The 8-frame history condition is built in two ways: 60% of samples use a
 728 standing cold-start pose, and 40% use noisy ground-truth history with Gaussian perturbations. This
 729 mixture trains the model to start from rest and to continue motion stably across sliding-window
 730 boundaries.

731 F Downstream Deployment: Detailed Tracker Specification

732 This appendix provides the complete formulation of the motion tracking policy used to validate the
 733 physical executability of trajectories generated by FlowOp.

734 F.1 Motion Tracking Formulation

735 **Observation Space.** The policy operates on

$$\mathbf{o}_t = \{\mathbf{p}_t, \mathbf{m}_t, \mathbf{z}_d\}, \quad (22)$$

736 consisting of proprioceptive robot states, kinematic motion references, and a compact motion latent
 737 token.

738 The proprioceptive observation is

$$\mathbf{p}_t = [\mathbf{g}_t^{\text{proj}}, \boldsymbol{\omega}_t, \mathbf{q}_t - \mathbf{q}_0, \dot{\mathbf{q}}_t, \mathbf{a}_{t-1}], \quad (23)$$

739 where $\mathbf{g}_t^{\text{proj}}$ denotes the gravity vector projected into the root frame, $\boldsymbol{\omega}_t$ is the root angular velocity,
740 $\mathbf{q}_t$ and $\dot{\mathbf{q}}_t$ are joint positions and velocities, $\mathbf{q}_0$ is the nominal joint configuration, and $\mathbf{a}_{t-1}$ is the
741 previous action.

742 The motion-reference observation is

$$\mathbf{m}_t = [\mathbf{v}_t^{\text{ref}}, \boldsymbol{\omega}_t^{\text{ref}}, \mathbf{e}_t^{\text{root}}, \mathbf{q}_t^{\text{ref}}], \quad (24)$$

743 where $\mathbf{v}_t^{\text{ref}}$ and $\boldsymbol{\omega}_t^{\text{ref}}$ denote target root linear and angular velocities, $\mathbf{q}_t^{\text{ref}}$ specifies target joint
744 positions, and $\mathbf{e}_t^{\text{root}}$ is a 6D root orientation error represented using Rot6D features.

745 The policy outputs target joint positions $\mathbf{a}_t \in \mathbb{R}^{29}$, which are tracked by a low-level PD controller.

746 **Motion-Latent Policy Architecture.** To improve temporal consistency and motion generalization,
747 the tracker adopts a shared encoder–quantizer–decoder architecture built around a compact motion
748 latent representation.

749 A motion encoder processes a future motion window $\mathbf{M}_{t:t+9}$ (10 frames at 50 Hz) and maps it
750 to a continuous latent embedding $\mathbf{z}_c \in \mathbb{R}^{256}$. This embedding is discretized using Finite Scalar
751 Quantization (FSQ) [27] into compact motion tokens $\mathbf{z}_d \in \{0, \dots, 7\}^8$, yielding a codebook of size
752 $8^8 \approx 16.7\text{M}$ discrete codes.

753 Two decoder branches consume $\mathbf{z}_d$:

- 754 • **Reconstruction decoder:** Reconstructs the future motion sequence $\hat{\mathbf{M}}_{t:t+9}$ for self-supervised
755 latent representation learning.
- 756 • **Action decoder:** Combines $\mathbf{z}_d$ with a short proprioceptive history (10 frames) to predict the control
757 action $\mathbf{a}_t$.

758 This latent conditioning stabilizes tracking across motions with substantially different dynamics and
759 contact patterns, enabling the same policy to handle diverse locomotion styles without per-motion
760 tuning.

761 F.2 Reward Function

762 **Tracking Objectives.** We employ eight tracking rewards that supervise kinematic alignment
763 between the executed motion and the reference trajectory. Each reward uses an exponential kernel
764 $r_i = w_i \exp(-\|\mathbf{x}_i - \mathbf{x}_i^{\text{ref}}\|^2 / \sigma_i^2)$, where w_i is the reward weight and σ_i is the kernel bandwidth. The
765 tracking terms are:

- 766 • Root position: $w=0.1, \sigma=0.30$ m
- 767 • Root orientation: $w=0.5, \sigma=0.40$ rad
- 768 • Root linear velocity: $w=1.0, \sigma=0.50$ m/s
- 769 • Root angular velocity: $w=1.0, \sigma=1.00$ rad/s
- 770 • Relative body position: $w=1.0, \sigma=0.30$ m
- 771 • Relative body orientation: $w=1.0, \sigma=0.40$ rad
- 772 • Body linear velocity: $w=1.0, \sigma=1.00$ m/s
- 773 • Body angular velocity: $w=1.0, \sigma=\sqrt{\pi}$ rad/s

774 **Regularization Terms.** Three regularization penalties are applied:

- 775 • Action smoothness: $w=-0.1 \cdot \|\mathbf{a}_t - \mathbf{a}_{t-1}\|^2$ (penalizes rapid changes)
- 776 • Joint limits: $w=-10 \cdot \sum_j \max(0, |\mathbf{q}_j - \mathbf{q}_j^{\text{default}}| - \mathbf{q}_j^{\text{limit}})^2$
- 777 • Undesired contact: $w=-0.1 \cdot \mathbb{I}[\text{non-foot contact}]$

778 All weights and bandwidths are fixed across all experiments without per-motion tuning, ensuring the
779 tracker remains general-purpose.

780 F.3 Domain Randomization

781 To improve sim-to-real transferability, we apply the following randomization during training:

782 Command-Level Perturbations.

- 783 • Target joint position noise: $\mathcal{U}(-0.01, +0.01)$ rad
- 784 • Target angular velocity noise: $\mathcal{U}(-0.2, +0.2)$ rad/s
- 785 • Target root rotation perturbation: $\mathcal{U}(-0.05, +0.05)$ rad (roll, pitch, yaw)
- 786 • Nominal joint configuration shift: $\mathcal{U}(-0.01, +0.01)$ rad

787 Physical Property Randomization.

- 788 • Center-of-mass offset: $x \sim \mathcal{U}(-0.5, +0.5)$ m, $y, z \sim \mathcal{U}(-0.1, +0.1)$ m
- 789 • Static friction coefficient: $\mu_s \sim \mathcal{U}(0.3, 1.6)$
- 790 • Dynamic friction coefficient: $\mu_d \sim \mathcal{U}(0.3, 1.2)$

791 **External Disturbances.** Random pushes are applied to the root body every 1–3 seconds:

- 792 • Linear impulse: $v_{xy} \sim \mathcal{U}(-0.5, +0.5)$ m/s, $v_z \sim \mathcal{U}(-0.2, +0.2)$ m/s
- 793 • Angular impulse: $\omega_{RP} \sim \mathcal{U}(-0.52, +0.52)$ rad/s, $\omega_Y \sim \mathcal{U}(-0.78, +0.78)$ rad/s

794 F.4 Training Procedure

795 **Pretraining Phase.** The tracker is first trained as a general-purpose motion imitation policy on a
796 large and diverse motion corpus consisting of:

- 797 • AMASS locomotion sequences: 12,500 clips
- 798 • Retargeted humanoid motions: 8,200 clips
- 799 • Synthetic motions (procedural gaits, transitions): 3,800 clips

800 Training uses PPO with 8 parallel environments, learning rate 3×10^{-4} , batch size 4096, and runs
801 for 30,000 policy updates (~ 10 hours on 8x RTX 4090).

802 **Fine-Tuning Phase.** After pretraining convergence, the policy is fine-tuned on trajectories generated
803 by FlowOp. Fine-tuning uses the same hyperparameters but with a reduced learning rate of 1×10^{-4}
804 and runs for 5,000 updates. This adaptation aligns the tracker with the motion distribution induced by
805 MA-A2R outputs while preserving the robustness acquired during large-scale pretraining.

806 F.5 Metrics

807 We evaluate tracking quality using:

- 808 • **Joint position error (JPE):** Mean L2 error between executed and reference joint angles (deg)
- 809 • **Root velocity error (RVE):** Mean L2 error in root linear velocity (m/s)
- 810 • **Success rate:** Fraction of rollouts completing without falls (within 30 seconds)

811 The resulting policy serves as the final deployment layer, validating whether FlowOp-generated
812 trajectories remain dynamically stable, physically trackable, and transferable to real-world humanoid
813 systems.

814 **G Additional Experimental Results**

815 **H Discussion**

816 **H.1 Limitations**

817 Although FlowOp substantially reduces the cross-morphology animation-to-robot transfer error on
818 AnimBot, several limitations remain. First, sceneFlow is a morphology-light motion descriptor, not a
819 fully morphology-blind one. Comparing the two main tables, MPJPE rises from 0.77 to 5.62. This
820 suggests that the model may still exploit residual morphology cues. Second, our experiments focus
821 on the 35-joint Unitree G1. FlowOp is designed to be K -agnostic, but this does not imply zero-shot
822 transfer to arbitrary robot bodies. Changing the target robot may still require a small amount of
823 target-side data for fine-tuning. Finally, the current method targets animation-to-humanoid transfer.
824 When the source motion has strongly non-humanoid topology, such as quadruped or multi-legged
825 motion, visual motion fields and target-skeleton binding alone cannot guarantee full executability.

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